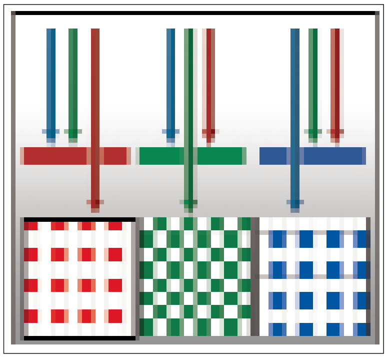
with the need for an ADC in many cases.

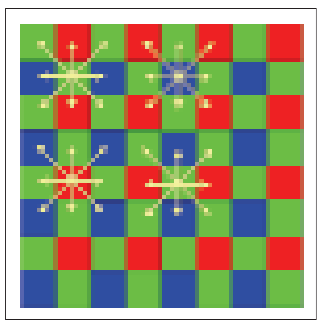
These different technologies have a number of advantages and disadvantages apparent from their functional peculiarities. Since each pixel on a CMOS sensor has a number of transistors neighbouring it the light sensitivity is reduced since with the same exposure some light will hit these dead zones i.e. the transistors. A CCD doesn't have this problem, but they consume a lot more power—in the range of 50 to a 100 times more than CMOS sensors.



How the filter works

For each image you click, there are millions of calculations that take place inside the camera. It's these calculations that allow the camera to interpolate, preview, capture, compress, filter, store and transfer the image as well as adjust settings, and allow manual tuning before taking a shot. The latest image processors are programmable which allows them to perform some advanced functions like in-camera photo editing, red eye removal, picture borders, panorama views and stitching, and even removing the blur caused by a shaky hand.

We spoke about each diode working in greyscale only. The next big step to an image is to add colours to it. This is where the Image Processor and the Image Sensor work together. In order to get colours into the image, the sensor must use filtering to look at the light (that they have captured) in its three primary colours. Once these three colours have been recorded, they are used in combination with each other to create all the colours. The three primary colours are also called RGB, where R is Red, G is Green and B is Blue. Nearly all cameras use what is called a Bayer Filter. This filter consists of a row of red and green blocks alternated with a row of blue and green blocks, something like a set of mosaic tiles. Note that the total number of individual blocks of each colour isn't identical and there are around two times more green blocks than any other colour simply because the eye is more sensitive to the colour green. Also note that the pixels on the filter are such that they can only capture one colour (that is, the colour corresponding to the colour of the filter placed directly above that pixel). Such a colour filter is placed over each pixel-capturing micro site and the image processor guesses the exact colour (read true colour) of a pixel by combining the actual colour that it captured corresponding to that pixel with the other two colours captured by the pixels surrounding it. If the raw output of a digital camera image sensor with a Bayer filter were to be viewed it would appear as a grid of red, green and blue pixels of varying intensities. There is then a process to break down this mosaic of estimated colours into another mosaic pattern with true colours. One major advantage is that a particular colour (of a particular pixel) can be used again by another pixel if required.



The bayer filter

Cool Fact

In 1860 James Clerk Maxwell created a colour photograph by using a black and white camera to take three photos of a tartan ribbon, and each photo was taken with a single colour filter in place (red, green and blue). The three black and white images were then projected onto a screen using three discrete projectors. When brought into alignment this formed a full colour photograph, and forms the basis for all cameras since.

Exposure and Focusing: Camera Optics

A digital camera has to control the amount of light reaching the Image Sensor, just like a film camera has to control exposure of the film. There are two components that work together to control the amount of light that is captured. They are the aperture and shutter speed. The Aperture is very simply the size of the opening in the camera that is the opening which allows light inside, and on to the image sensor. Mostly digital cameras have automatic apertures, although a manual setting is provided for enthusiasts and professionals to have more control over the final image. The shutter speed controls the duration for which light is allowed to pass through the aperture. Digital cameras can have both analog and digital shutters.

The lens of a digital camera is very similar to the lens on a conventional film camera. The role of the lens too remains the same—to control how light is focussed on the image sensor for which the lens actually moves. All digital cameras use autofocus—a feature where the lenses automatically adjust to focus on the subject. Where there are more than one subjects advanced autofocus methods may be used or manual focus is also available on digital SLR cameras.

The last part of the camera optics is the focal length. The focal length is defined as the distance between the lens and the surface of the sensor. But whereas for a film camera the size of the film is fixed at 35 mm, for a digital camera the size of the image sensor changes from manufacturer to manufacturer and model to model. In most digital cameras the size of this sensor is smaller than the size of a 35-mm film, so in order to project the image on this sensor the focal length also has to be reduced in proportion. With a zoom lens (optical zoom) the focal length is actually changed so that the subject appears to be much closer.

In Conclusion

Think of your digital camera as a miniature computer. As camera resolutions increase and image sensors have evolved, digital photography has become good enough to become a household commodity, and the focus has now shifted to features and usability. Features like face recognition, on-the-fly red eye removal and even colour and tone adjustment in an image, grant more control over image settings even as they complicate things (unnecessarily for some) over the default Auto mode. With the latest consumer cameras upping the MP rating and optical zooms, and a host of other specifications we've come a long way from black and white giving way to colour. Mr Maxwell would be pleased...

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